Layer Composition and Predictive Stability in Transformers: Contextual Reachability and Distributed Head Contributions

Paper 0.5 — Residual / Layer-Composition Research Line

Integrated controlled report — August 2026

Abstract

We study three linked phenomena: how depth changes the stability and correctness of a response under irrelevant input variation, how the self-attention (SA) window limits transport of predictive context, and how heads contribute to stability and class discrimination. The primary instruments are custom causal Transformers trained on known synthetic distributions. Dataset V2 eliminates singleton and proper-subset shortcuts and crosses six balanced predictive formats with four nuisance levels. Across three seeds, mean held-out generator accuracy is 0.885–0.913, 89.5% of identities reach stable top-1 behavior, and 70.3% of nested cases reach a stable long-rule override. Nuisance count predicts later stabilization descriptively ($R^2 = 0.354$), whereas pattern length and span do not. A new span–distance–window experiment finds that generic inference-time restriction harms full-trained models, but selectively blocking nuisance keys preserves perfect accuracy, increases margin by 0.23 logits, and reduces margin variance by 40.5%. Locally trained restricted models recover perfect accuracy with lower variance. SA and FF both change variance substantially, and median cross-layer repair fraction is 0.812; SA is therefore a transport mechanism, not a unique noise source. Cumulative Jacobian–vector products are directionally informative but not accurate global surrogates. Head interventions show distributed non-additive effects. The supported theory is composed, anisotropic, state-dependent predictive refinement—not universal denoising, a sufficient $Lw \geq s$ law, generic "less is more," or globally linear dynamics.

1 The question and the three phenomena

Let an input be $x = (g, d)$, where g is the predictive structure and d is constructed nuisance, with

$$Y \perp d \mid g. \tag{1}$$

We ask whether depth organizes the many surface realizations (g, d_i) into a stable predictive class while retaining distinctions between different g. This system-level question is investigated in dependency order:

1. *layer composition*: does stability improve, and is the trajectory monotone?
2. *SA transport*: can decisive context reach the prediction position through a finite window?
3. *heads*: which parallel SA channels stabilize a class or distinguish classes?

Continuous learning is not the experimental objective. Pretrained and prior mechanistic results are retained as secondary external-validity evidence in the appendices.

2 A pedagogical theory of predictive ensembles

For a pre-norm block,

$$r'_\ell = r_\ell + A_\ell(r_\ell), \qquad r_{\ell+1} = r'_\ell + M_\ell(r'_\ell). \tag{2}$$

The key point is that layer $\ell + 1$ acts on the state produced by layer ℓ; layers are composed transformations, not independent estimates. For family g, imagine a cloud

$$\mathcal{E}_\ell(g) = \{r_\ell(g, d_i)\}_{i=1}^{m}. \tag{3}$$

Its centroid $\mu_{\ell,g}$ describes the common state and covariance $\Sigma_{\ell,g}$ describes realization-sensitive spread. Applying the unembedding at every depth gives distributions $p_\ell(\cdot \mid g, d)$. We measure

$$D_{\text{within}}(\ell) = \mathbb{E}_g \mathbb{E}_d [JS(p_\ell(\cdot \mid g, d), \bar{p}_{\ell,g})], \tag{4}$$

$$D_{\text{between}}(\ell) = \mathbb{E}_{g \neq g'} [JS(\bar{p}_{\ell,g}, \bar{p}_{\ell,g'})], \tag{5}$$

$$R_\ell = \frac{D_{\text{between}}(\ell)}{D_{\text{within}}(\ell) + \epsilon}. \tag{6}$$

A useful layer trajectory lowers within-family fluctuation without erasing target signal or between-family separation. Large R_ℓ is not self-interpreting when its denominator is nearly zero, so we always report accuracy, target probability, and numerator/denominator with it.

2.1 Correctness-aware order parameters

Let $z_\ell(Y)$ be the correct-target logit and let $z_\ell(c^*)$ be the strongest competing logit. Define the per-realization margin

$$m_\ell(g, d) = z_\ell(Y) - z_\ell(c^*), \tag{7}$$

which is positive exactly when the target is top-1. Across nuisance realizations, the signal and fluctuation are

$$S_\ell = \mathbb{E}_d[m_\ell(g, d)], \qquad N_\ell^2 = \mathrm{Var}_d[m_\ell(g, d)], \tag{8}$$

$$\Gamma_\ell = \frac{S_\ell}{\sqrt{N_\ell^2 + \epsilon}}, \qquad \Gamma_\ell^{\text{robust}} = \frac{\text{median}_d\, m_\ell}{1.4826\, \text{MAD}_d(m_\ell) + \epsilon}. \tag{9}$$

We always show S_ℓ and N_ℓ^2 beside Γ_ℓ: a large ratio can otherwise arise from a collapsed denominator. Correctness and class organization are complementary: Γ_ℓ asks whether the externally specified answer wins reliably, whereas R_ℓ asks whether families separate. In representation space we use

$$\chi_\ell = \frac{q_\ell}{M_\ell^{\text{class}} + \epsilon}, \quad q_\ell = \mathbb{E}_g \operatorname{Tr} \Sigma_{\ell,g}, \quad M_\ell^{\text{class}} = \mathbb{E}_{g \neq g'} \|\mu_{\ell,g} - \mu_{\ell,g'}\|^2. \tag{10}$$

The first top-1 depth is the first boundary where $m_\ell > 0$; stable depth is the first boundary after which it remains positive. Never-correct trajectories use a missing sentinel, never the final layer.

For boundary-to-boundary progress $a_{\ell i} = m_{\ell+1,i} - m_{\ell,i}$, write $a_{\ell i} = \mu_\ell^\Delta + \epsilon_{\ell i}^\Delta$. Then

$$S_L^{\text{cum}} = \sum_\ell \mu_\ell^\Delta, \tag{11}$$

$$(N_L^{\text{cum}})^2 = \sum_\ell (\sigma_\ell^\Delta)^2 + 2 \sum_{\ell < k} \mathrm{Cov}(\epsilon_\ell^\Delta, \epsilon_k^\Delta). \tag{12}$$

This decomposition makes repair visible: negative cross-layer covariance means that fluctuations introduced at one boundary tend to be counteracted later.

2.2 Why non-monotonicity is expected

For a perturbation δr_ℓ, linearize one block:

$$\delta r_{\ell+1} \approx J_\ell \delta r_\ell, \qquad J_\ell = I + \frac{\partial F_\ell}{\partial r}. \tag{13}$$

After L blocks the relevant map is the product $J_{L-1} \cdots J_0$. One factor may expand a nuisance direction and a later factor may rotate or contract it. Hence final contraction does not imply contraction at every layer. We test the approximation with Jacobian–vector products (JVPs), never by constructing a dense Jacobian.

2.3 Context as a causal cone

With a local window w, one layer can move information at most w positions in the mask graph. Thus $Lw \geq s$ is a necessary reachability condition for a source s positions away in the simple mask used here. It is not sufficient: attention weights, width, optimization, and intermediate coding can still prevent useful transport.

2.4 Heads as vector-field components

Write $A_\ell(r) = \sum_h A_{\ell h}(r)$. For head h, define

$$U^{\mathrm{stab}}_{\ell h} = D^{(-h)}_{\mathrm{within}} - D^{\mathrm{intact}}_{\mathrm{within}}, \tag{14}$$

$$U^{\mathrm{disc}}_{\ell h} = D^{\mathrm{intact}}_{\mathrm{between}} - D^{(-h)}_{\mathrm{between}}. \tag{15}$$

For two heads, $\Gamma(h,k) = U(h,k) - U(h) - U(k)$ distinguishes synergy ($\Gamma > 0$) from redundancy or compensation ($\Gamma < 0$). These are causal, task-conditioned quantities; an attractive attention motif is not a substitute.

3 Controlled methods

The generator records predictive family, held-out surface identity, target, pattern tokens/length/span, nuisance tokens/count/type/seed, answer-changing status, continuation entropy, training frequency, split, and generator seed. Group 1 uses contiguous n-grams, skip-grams, and binary functors: nine families, three model seeds, nuisance counts 0/4/8, twelve realizations, width 48, four layers, and four heads. At the initial pre-SA state and every post-SA/post-block boundary we retain target logit/probability/rank/margin, strongest competitor, top-1 result, entropy, distribution hash, residual covariance, first/stable decision depth, reversals, layer-update signal/noise, and cross-layer covariance. Group 2 uses eight (s, w, L) cells, four predictive classes, an exactly tested local mask, and a separately labelled local-pattern task. Group 3 uses a trained four-layer/four-head full-attention model and evaluates every head with zero ablation, batch-mean replacement, same-family replacement, mismatched replacement, and every within-layer pair. Manifests store resolved configs, seeds, data identities, and checkpoint hashes. Family/seed bootstrap is used for the confirmatory Group 1 contrasts; head scans are descriptive and no row-level significance is asserted.

Dataset V2 strengthens the first group. Tokens are partitioned into role vocabularies but nuisance uses the same vocabulary as predictive tokens, preventing a trivial vocabulary filter. Balanced pair, three-token, sparse-pair, functor, short nested, and long nested rules have target entropy 2 bits, maximum singleton and proper-subset target mutual information 0 bits, and full-pattern target information 2 bits. Train and test surface identities are disjoint; target counts are exactly balanced; filler, position, nuisance, and competing-count checks pass. The confirmatory matrix has 11,696 examples per split, nuisance levels N0/N2/N3/N4, counts 0/2/4, aligned and randomized positions, and three independently initialized four-layer models. Every boundary records correctness, entropy, JS, representation covariance/effective rank, margin SNR, and decision/override timing.

Part I

I. Layer composition and response stability

4 No-shortcut Dataset V2 and nested override

All six balanced training families learn well above four-way chance over the complete held-out nuisance matrix. Mean accuracies are 0.913 for pairs, 0.885 for three-token sums, 0.908 for sparse pairs, 0.913 for functors, 0.890 for short nested rules, and 0.887 for long nested rules. The worst generator–seed–nuisance cell is 0.646. This minimum and the seed means (0.718, 0.993, and 0.994) reveal material optimization variability; we retain it rather than presenting only a pooled success.

Stable top-1 behavior occurs for 89.5% of held-out identities. Among nested long-rule cases, 70.3% eventually prefer the correct long target over the competing short-rule target and retain that preference. Thus the network often performs contextual override rather than merely repeating a locally valid suffix rule. The result is measurable but not universal.

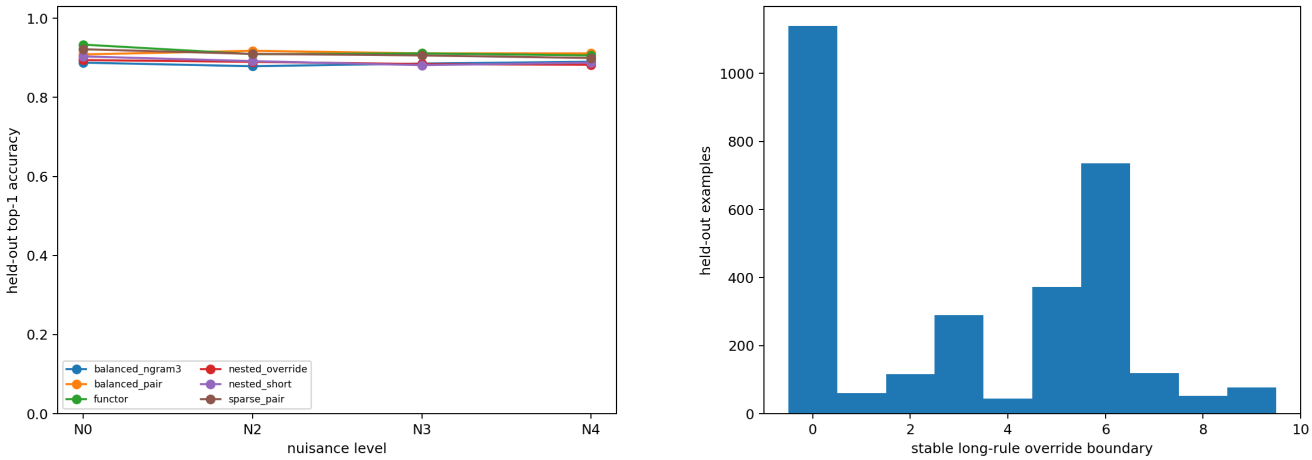

Figure 1: Dataset V2 held-out competence across nuisance levels and the boundary at which the longer nested rule stably overrides its valid short suffix.

Linear descriptive fits explain essentially none of stable-depth variation from pattern length ($R^2 = 0.0025$) or dependency span ($R^2 = 0.0004$). Nuisance count explains 35.4%, with an estimated delay of 2.57 sublayer boundaries per added nuisance token in this narrow design. This is evidence for nuisance-dependent computational delay, not a universal exponent: the response saturates at the finite model depth, the predictors are crossed imperfectly, and one seed is much weaker.

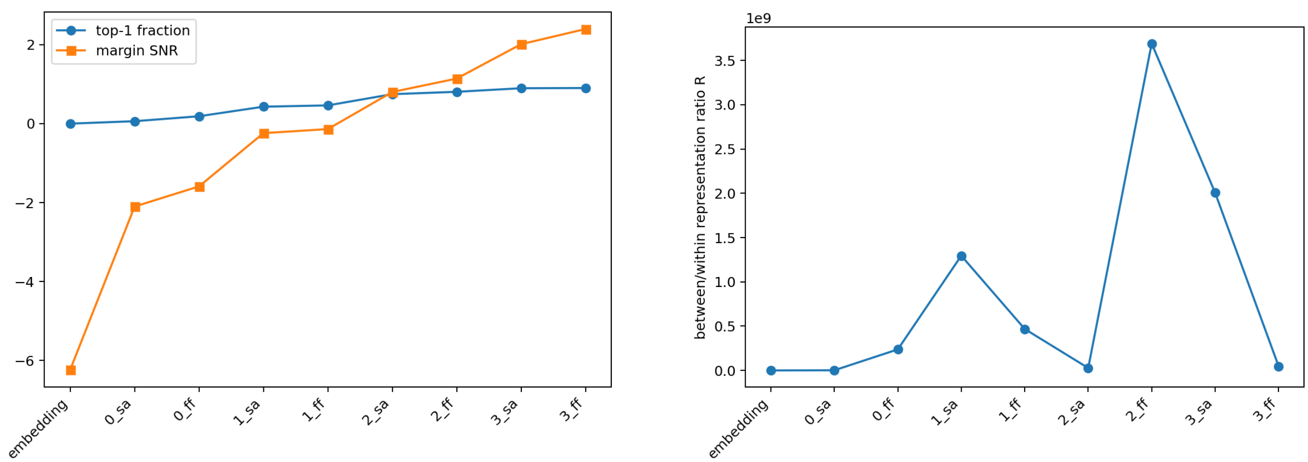

Figure 2: Correctness/SNR and representation separation-to-within-family variation across Dataset V2 depth.

5 Pilot reproduction and generalization

The prior fixed-depth result reproduces exactly: under pattern evidence and four distractors, target-probability variance is $0.0347 \rightarrow 0.0106 \rightarrow 0.0109$, JS dispersion is $0.0650 \rightarrow 0.0307 \rightarrow 0.0106$ bits, and within-example entropy is $1.326 \rightarrow 2.094 \rightarrow 2.277$ bits. Stable response across realizations is therefore different from low entropy within one realization.

In the new matrix, all three generators achieve 100% final accuracy with eight nuisances. First-to-last within-JS changes are -0.0244 bits for contiguous n-grams (family/seed CI $[-0.0414, -0.0098]$), -0.0415 for skip-grams ($[-0.0791, -0.0129]$), and -0.0402 for binary functors ($[-0.0738, -0.0143]$). Stability generalizes beyond contiguous n-grams without sacrificing target signal.

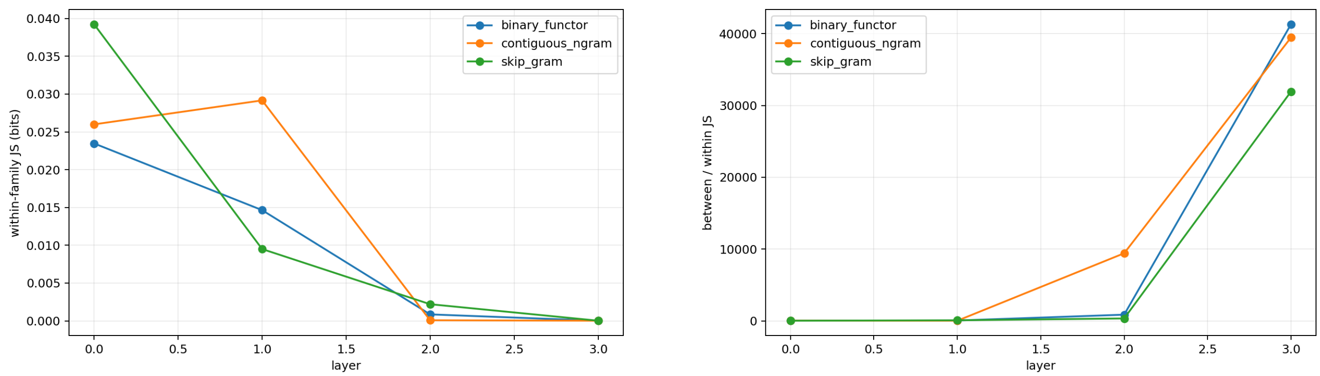

Figure 3: Within-family JS and the separation-to-fluctuation ratio across depth under eight nuisances.

6 Correct signal grows faster than margin fluctuation

At eight nuisances, the aggregate mean correct-target margin changes from -32.02 logits initially to $+10.51$ finally, mean target probability from 1.4×10^{-10} to 0.9998, and top-1 fraction from 0 to 1. Each generator independently changes from negative to positive margin. This rules out the “stable but uniformly wrong” failure mode.

The stronger absolute-contraction hypothesis is nevertheless false here. Margin variance starts at zero because the aligned prediction-position state is initially identical across nuisance realizations; it finishes at 1.89. Final median margin SNR is $+10.24$. Thus reliability comes from roughly 42.5 logits of signal growth outpacing fluctuation growth, not from an SNR denominator collapse. Prediction-space JS can contract even while logit-margin variance expands because the softmax geometry and the mean decision direction both change.

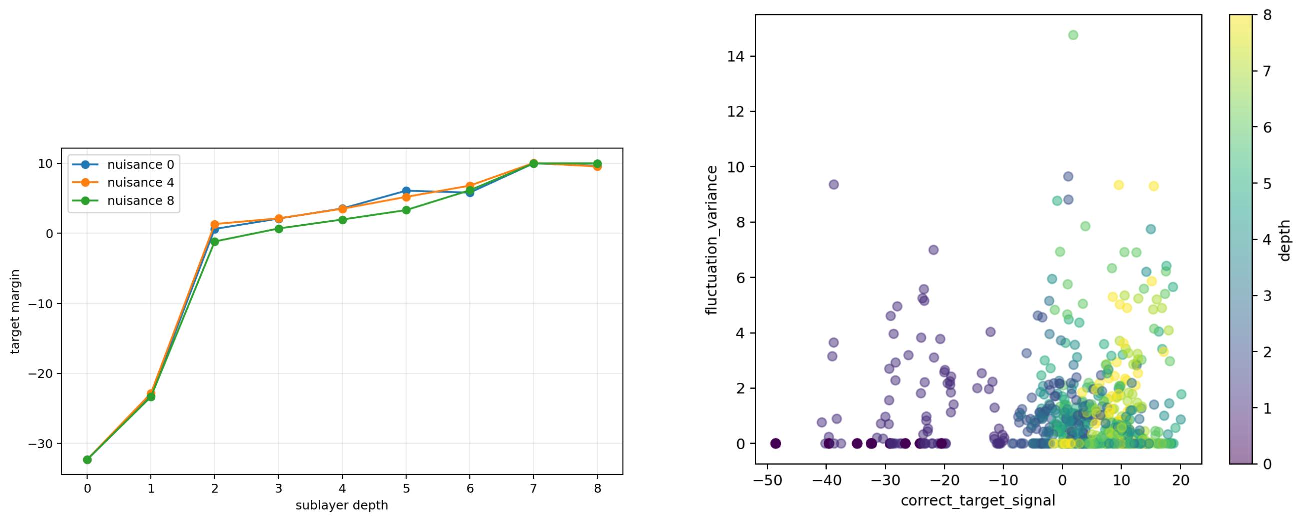

Figure 4: Correct-target margin across sublayer depth and the signal–fluctuation phase plane. The trajectory moves from confidently wrong and identical to confidently correct with finite nuisance variation.

7 Decision formation, update covariance, and scaling non-results

Mean first/stable top-1 depths are 3.00/3.40 boundaries at nuisance 0, 3.08/3.48 at nuisance 4, and 3.41/3.88 at nuisance 8. Mean settling delays are 0.48, 0.40, and 0.47; mean correctness reversals remain 1.37–1.52. Nuisance modestly delays decision formation, but a five-form comparison finds no scaling law: the best stable-depth fit has $R^2 = 0.013$ and held-out error 1.65 boundaries. Positive-margin SNR fits explain at most 1.0% of variance; logarithmic and linear BIC differ by less than one. A power law is unsupported.

Mean boundary update is +5.12 logits and 73.7% of family/seed/noise transition cells have positive mean progress. Median cumulative signal is +40.39 logits. Median summed marginal update variance is 3.55, but median summed off-diagonal covariance is -2.12, leaving actual cumulative fluctuation variance 0.51. Later updates therefore cancel a substantial part of earlier realization-specific fluctuation.

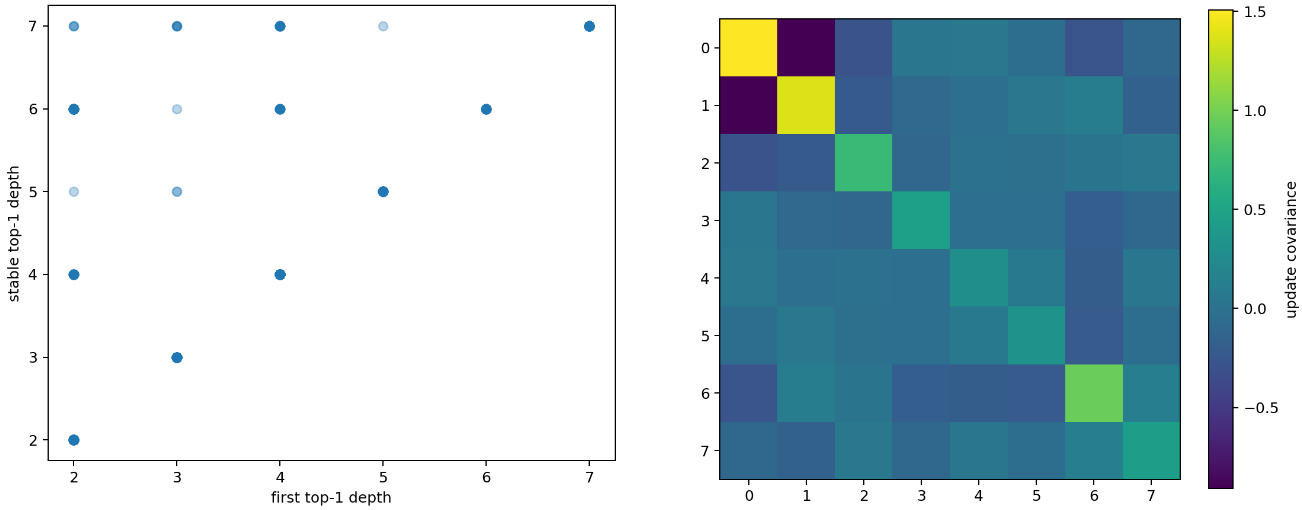

Figure 5: First versus stable decision boundaries and cross-boundary margin-update covariance. Negative off-diagonal structure implements fluctuation cancellation/repair.

Entropy remains independent of correctness: its correlation with aggregate margin is only 0.081. It can increase while margin crosses zero and while within-family JS contracts.

8 Non-monotone composition and local linearization

Only 7/27 family–seed trajectories contract monotonically; 11 expand then contract, five oscillate, three contract late, and one does not contract. This favors cumulative nonlinear composition over a layerwise denoising story.

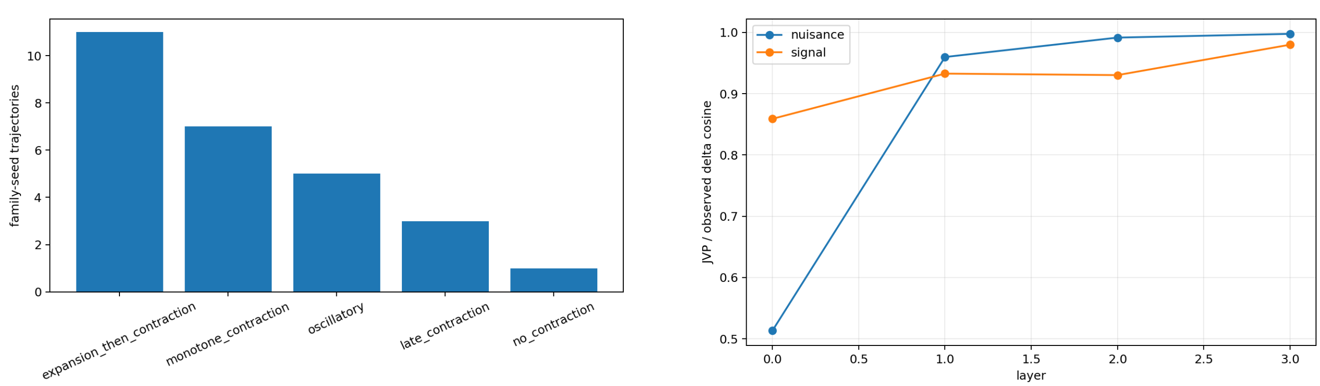

Figure 6: Trajectory classes and local JVP agreement. Median JVP/observed cosine is 0.984 for nuisance and 0.928 for signal directions; prediction-space relative errors are 0.179 and 0.311.

The Jacobian describes small nuisance propagation better than the larger answer-changing displacement, but neither local statistic determines the product of later transformations.

Dataset V2 directly tests the product. For a pair difference δr_a, we repeatedly apply JVPs along the reference trajectory, $\widehat{\delta r_b} = J_{b-1} \cdots J_a \delta r_a$, and compare it with the observed paired difference. Across 85 behaviorally filtered controlled pairs, mean cosine is 0.782 for one block, 0.552 for two, and 0.335 for four; relative error rises from 0.624 to 0.880 and 1.047. At four blocks, nuisance directions retain higher cosine than answer-changing signal controls

(0.476 versus 0.227), and piecewise local re-linearization has substantially lower error than the frozen product. The product therefore captures local direction and accumulated nonlinearity, not a faithful global linear replacement.

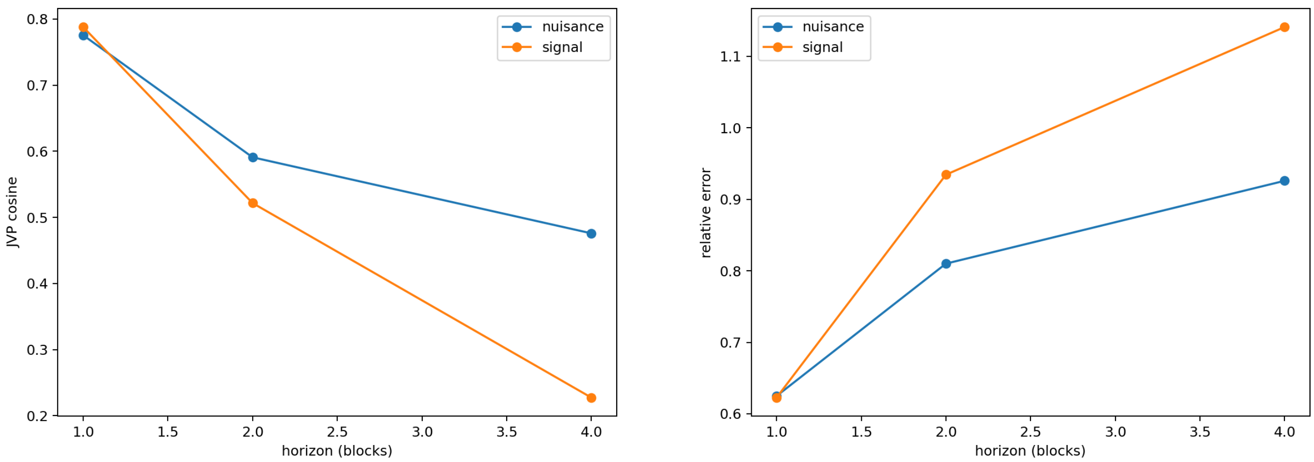

Figure 7: Controlled cumulative-JVP directional agreement and relative error as the composition horizon grows.

Part II

II. Self-attention window and contextual transport

9 Reachability is necessary, not sufficient

The mask tests verify exact graph reachability. Unreachable long-range cells remain near four-way chance: accuracies are 0.188 for $(s, w, L) = (2, 1, 1)$, 0.234 for $(4, 1, 2)$, and 0.234 for $(8, 2, 2)$. Reachable $(2, 1, 2)$ reaches 0.797; $(4, 1, 4)$, $(8, 4, 2)$, and $(8, \text{full}, 2)$ reach 1.0. Yet reachable $(8, 2, 4)$ reaches only 0.078. The causal cone is a necessary structural bound, not a sufficient learning law.

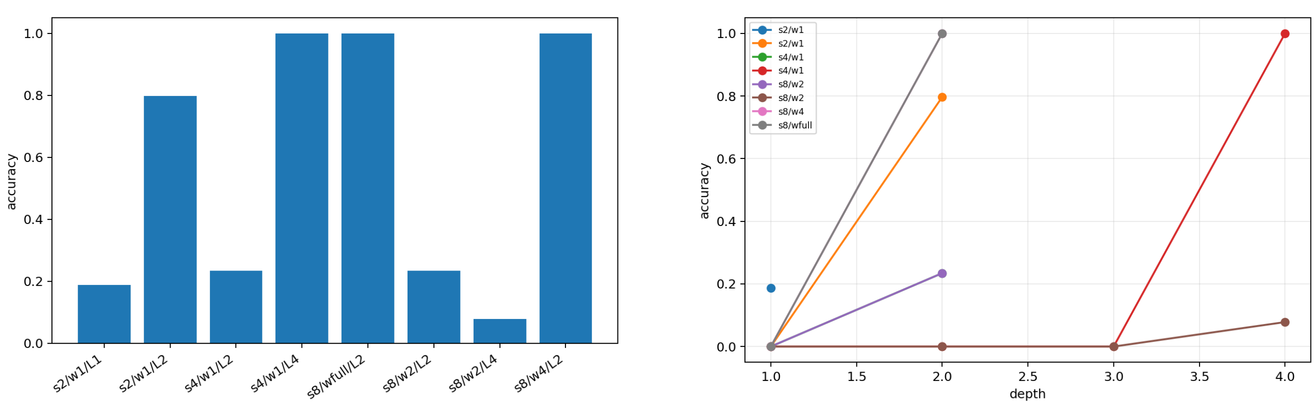

Figure 8: Final accuracy by span/window/depth cell and layerwise functional arrival.

Local-pattern accuracy is 0.75–0.953 in most cells, showing substantial robustness when the decisive token is retained, but it is not uniformly at ceiling. Dataset V2 supplies the separate nested-override trajectory above; it does not repair this window sweep's single reachable optimization failure.

10 Experiment 2.5: information exposure and variance attribution

We next cross predictive span $s_G \in \{2, 8, 16\}$, nuisance distance $s_N \in \{2, 4, 8, 16, 32\}$, effective window $w \in \{1, 2, 4, 8, 16, \text{full}\}$, depths 2/4/6, two seeds, and three nuisance types. The target is the balanced rule $Y = (A + B) \bmod 4$, for which neither token is predictive alone. N1 uses same-vocabulary random tokens, N2 partial predictive fragments, and N3 complete competing patterns. Full-attention models are evaluated under all truncated windows without retraining; three locally trained window controls distinguish immediate exposure from adaptation.

The first result falsifies a generic "less is more" account. Full-trained/full-evaluated models achieve 0.9995 accuracy with final margin variance 1.890. Merely truncating these models often destroys the learned routing: the graphically signal-reachable/nuisance-unreachable regime reaches only 0.411 accuracy. In contrast, locally trained restricted models reach 1.000 accuracy with mean variance 1.110. Thus $Lw \geq s_G$ states possibility, not learned functional transport, and training regime cannot be ignored.

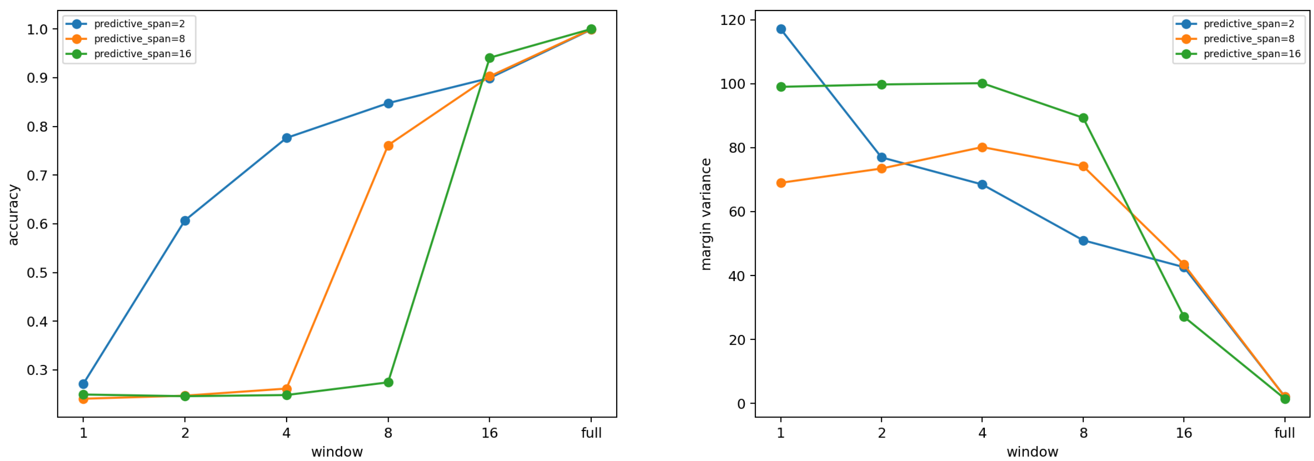

Figure 9: Full-trained models under inference-time window restriction. Accuracy and margin variance improve toward full exposure; reachability alone does not preserve the learned computation.

Selective attention masking supplies the cleaner causal test. Blocking only nuisance key positions while retaining their embeddings preserves 1.000 accuracy, increases mean target margin by 0.229 logits, and reduces nuisance-conditioned margin variance from 1.864 to 1.109 (40.5%). Blocking two matched random positions changes margin by only -0.034 logits, whereas blocking the two predictive positions reduces accuracy to 0.249 and raises variance to 67.40. SA therefore transports nuisance values that measurably perturb the decision variable, but excluding all distant context is not equivalent to excluding the causally irrelevant positions.

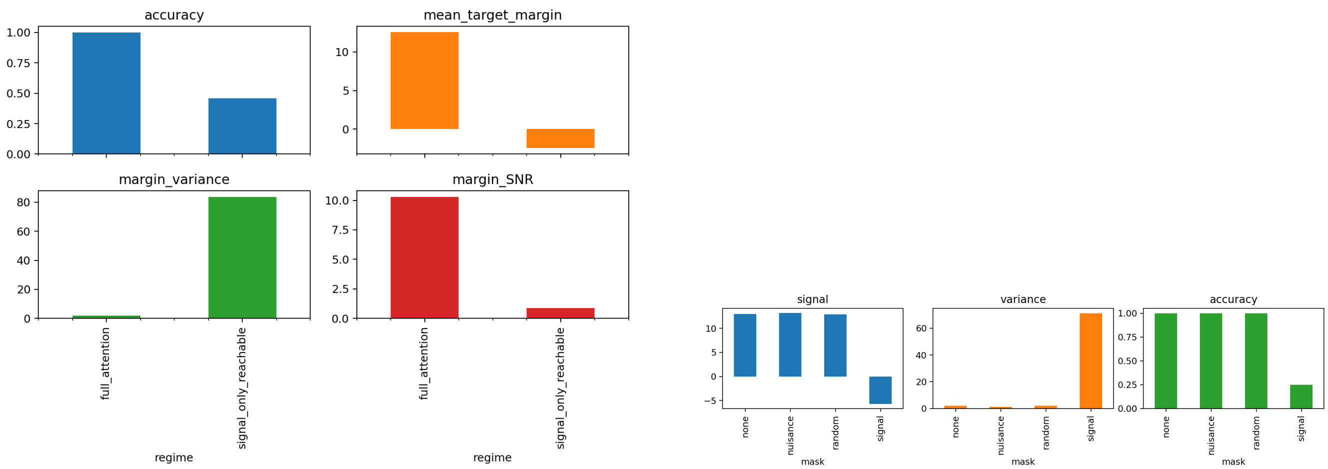

Figure 10: Left: graph-defined signal-only reachability does not match full-attention competence without local-window training. Right: selective nuisance removal lowers variance without sacrificing signal, while signal removal collapses performance.

Variance attribution also rejects the claim that SA is the sole injector. Across the full intervention grid, mean sublayer changes are +4.81 for SA and +4.49 for FF, while normalization is associated with −0.65; these averages are dominated by failing truncated evaluations and must not be read as pure nuisance import. In the nonlinear descriptive model (architecture-cell holdout $R^2 = 0.959$), window, signal magnitude, layer, normalization change, and signal attention mass rank above nuisance attention mass and distance. Mean off-diagonal update covariance is negative; 86.8% of valid cells have positive repair and the median repair fraction is 0.812. The supported mechanism is joint: SA expands accessible information, nonlinear SA/FF transformations amplify or rotate it, candidate competition changes the boundary, and later layers cancel much of the realization-specific deviation.

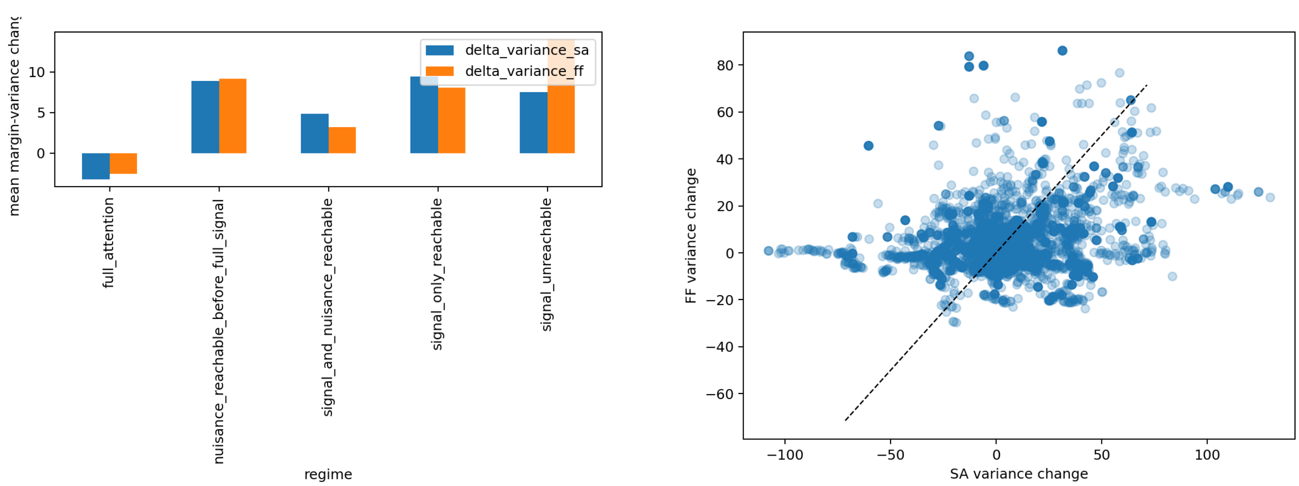

Figure 11: Mean sublayer variance changes by reachability regime and cellwise SA versus FF changes. SA is important, but not uniquely dominant, and the sign depends on learned competence.

The bridge to bounded materialization is correspondingly narrow. Selective exclusion of known irrelevant context can improve SNR, but indiscriminate restriction can erase the learned signal path. “Less” helps only when the budget removes nuisance while respecting the model’s trained routing.

Part III

III. Heads and the SA contribution

11 Replacement, specialization, and interaction

Across 16 layer/head cells, mean target-log-probability drops are 0.0316 nats for zero ablation, 0.0108 for mean replacement, 0.0006 for equivalent-family replacement, and 0.0194 for mismatched replacement. Zero ablation increases within-family JS by 4.0×10^{-5} bits and reduces between-family JS by 3.93×10^{-4} on average. Mismatch produces the largest discrimination damage (1.44×10^{-3} bits). Stability and discrimination are therefore related but non-identical head roles.

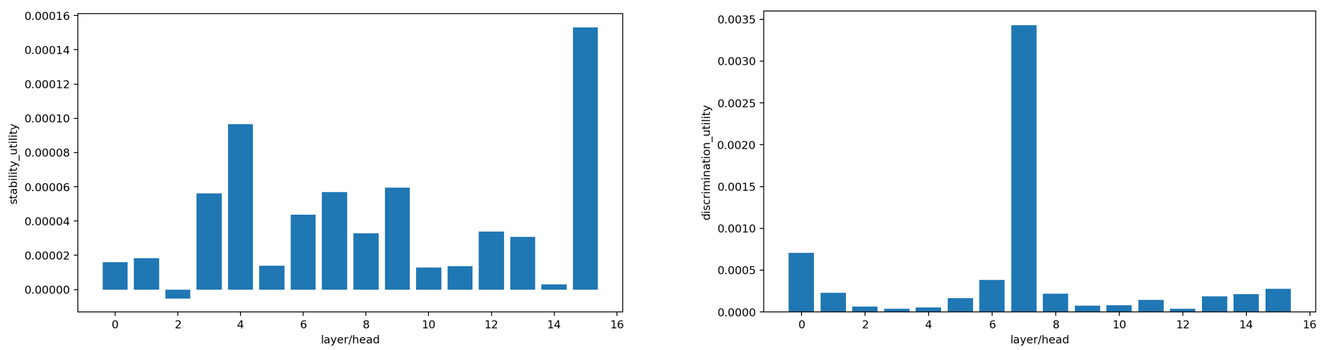

Figure 12: Causal stability and discrimination utilities for all layer/head cells.

Pair interactions range from -0.0116 to $+0.0187$ nats, with median $+0.0050$, so individual utilities do not add. Median correlation of head-output norms is 0.002, rejecting a simple globally correlated-channel account without establishing independence of the full vectors. Motif specificity is again unrelated to causal utility ($\rho = -0.0059$ across 16 cells), closely reproducing the prior -0.009 null.

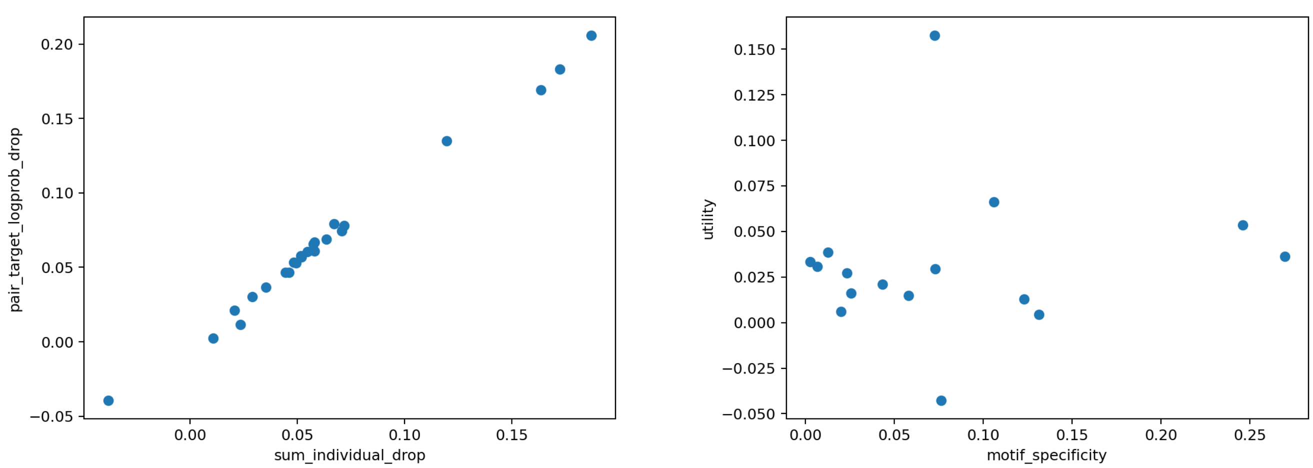

Figure 13: Non-additive pair effects and the repeated motif–causal null.

12 Integrated interpretation

The statistical view is supported in relative form: predictive-family JS contracts, correct margin becomes strongly positive, and signal grows faster than absolute margin fluctuation. The dynamical view is supported descriptively: decisions reverse, trajectories are usually non-monotone, cross-layer fluctuations anticorrelate, and cumulative JVP fidelity decreases with horizon. The information-theoretic premise is now directly checked at the data level: singleton and proper-subset target MI vanish while the full pattern carries 2 bits; representation-space mutual information is still not estimated. The statistical-mechanical view is partial: correctness signal, fluctuation, R, χ, and covariance are measured, but neither a universal scaling law nor a finite-size phase transition is supported.

The most economical account is *state-dependent anisotropic refinement*. Layers transport, rotate, expand, and contract different perturbations; SA reachability constrains which distinctions can enter the prediction position, while training determines whether reachable paths are useful. Selective masking shows that nuisance access contributes causally to variance, but the window surface and attribution model show comparable FF changes, normalization effects, competition, and substantial later repair. Heads supply parallel, interacting vector-field components. No single layer or head need be a complete estimator, and final stabilization need not be monotone.

13 Limitations and open tests

Dataset V2 uses three seeds but only one architecture scale and four blocks; it cannot identify universal exponents. One seed is materially weaker, and nuisance count is not orthogonal to every surface statistic. Zero nuisance has identical realizations and hence exactly zero fluctuation, producing extreme SNR values that are retained but excluded from positive-margin scaling fits. Experiment 2.5 uses two seeds, three depths, and a sparse rather than full Cartesian grid. Its nuisance-key intervention is stronger than its descriptive attribution model, but the synthetic positions are known exactly and do not address how nuisance should be identified in natural text. Group 2 contains a reachable optimization failure. Group 3 has one seed/model, no multi-seed head-role reproducibility, and only one head-count point. Huge R values from near-zero denominators are not treated as precise effect sizes.

A Prior controlled component and trajectory experiments

The preserved earlier experiment contains 21,630 atlas rows, 1,920 component rows, 1,920 motif/control rows, 2,880 entropy rows, 3,840 equivalence patches, and dense checkpoints. Familiar deterministic mappings are FFN-heavier under zero ablation (0.478 versus 0.301 nats) but not matched replacement; override/repair is SA-heavier (1.497 versus 0.261). Equivalent update replacement changes outputs less than nonequivalent replacement (layer-0 SA JS 0.028 versus 0.436 bits). Diagnostic logit progress often disagrees with causal magnitude, and controlled frequency/mechanism regression has only $R^2 = 0.073$. These results motivate distributed computation but are secondary to the three investigations.

B Pretrained external validity and manifold control

The new cumulative-Jacobian pilot precedes the older intervention evidence because it tests the same mathematical object as the controlled theory. Behavioral filtering retains 10 of 12 equal-length prompt pairs across pinned final Pythia-70M and Qwen3-0.6B. "Nuisance" here means only unchanged final top-1; signal controls change top-1. Mean cumulative directional cosine is 0.768, 0.733, and 0.702 over one, two, and four blocks. Directional correspondence therefore survives modest composition in both architectures. This is weak external validity, not a reconstruction theorem: relative-norm error exceeds one in several model/direction cells, Qwen nuisance piecewise error does not improve on its frozen path, and neither the true pretraining frequency nor the true predictive family is known.

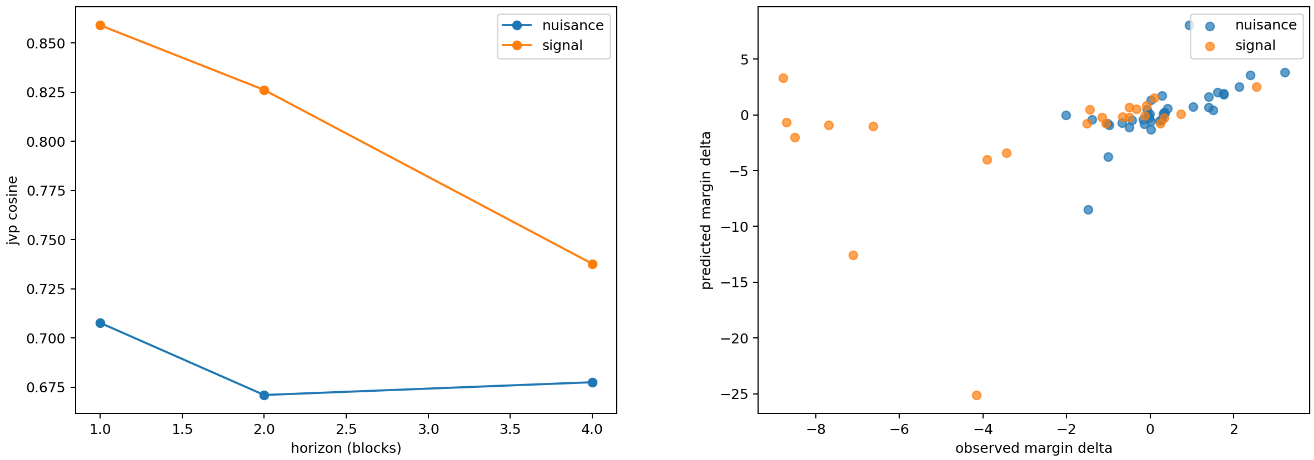

Figure 14: Behaviorally filtered pretrained cumulative-JVP direction and decision-margin projection. These are architectural diagnostics, not corpus-level equivalence claims.

The earlier four-family matrix found equivalent-minus-nonequivalent advantages of 1.047 nats for Pythia step 1,000, 1.206 for final Pythia, and 0.599 for Qwen, but no reliable random-initialization effect. The expanded 32-family matrix found fully-nonequivalent advantages of 2.552 nats for final Pythia (CI $[1.991, 3.070]$) and 0.763 for Qwen ($[0.623, 0.899]$), with syntax mismatch more harmful than semantic mismatch. This evidence is sharply qualified by behavioral competence (1.6% for Pythia versus 68.8% for Qwen) and an on-manifold falsification: Qwen's residual-nearest nonequivalent donor costs -0.022 nats (CI $[-0.082, 0.023]$), essentially identical to equivalent replacement. Rank-1 removal remains harmful (Pythia -1.019, Qwen -0.837 nats), while addition is harmful. Common subspaces are contributory and state-dependent, but functional labels have not been isolated from residual proximity.

C Override, training, and variance controls

Answer-changing contexts are never labelled nuisance. In the legacy grid their JS contraction is -0.0328 bits (CI $[-0.0543, -0.0141]$), weaker than irrelevant-nuisance contraction -0.0474 ($[-0.0778, -0.0217]$). Random checkpoints have low dispersion only because target signal is absent. Training-time SA/FF trajectories are non-monotone, and stable attention motifs do not predict causal SA utility. These negative controls constrain the theory rather than being discarded.

D Relationship to later continual-learning work

Only here, after the mechanistic results, do we state the downstream implication. Stable predictive equivalence could become a diagnostic precursor for future adaptation, but Paper 0.5 neither implements nor validates continual learning. Common-direction addition is not beneficial, on-manifold controls weaken label-specific transfer, and the separate natural held-out utility gate remains negative. Persistent prototypes or adapters therefore remain blocked pending matched-budget future utility, override preservation, contamination controls, and rollback.

E Reproducibility

```
PYTHONPATH=src python -m cl.experiments.paper05_layer_composition
  --group 1 --config configs/paper05/group1.json --device cpu
PYTHONPATH=src python -m cl.experiments.paper05_layer_composition
  --group 2 --config configs/paper05/group2.json --device cpu
PYTHONPATH=src python -m cl.experiments.paper05_layer_composition
  --group 3 --config configs/paper05/group3.json --device cpu
PYTHONPATH=src python -m cl.analysis.paper05_layer_composition
PYTHONPATH=src python -m cl.experiments.paper05_correctness
PYTHONPATH=src python -m cl.experiments.paper05_dataset_v2
```

```
PYTHONPATH=src python -m cl.experiments.paper05_dataset_v2_train
PYTHONPATH=src python -m cl.experiments.paper05_dataset_v2_train --evaluate-only
PYTHONPATH=src python -m cl.analysis.paper05_dataset_v2
PYTHONPATH=src python -m cl.experiments.paper05_dataset_v2_jacobian
PYTHONPATH=src python -m cl.experiments.paper05_group25_sa_variance
PYTHONPATH=src python -m cl.analysis.paper05_group25_sa_variance
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 PYTHONPATH=src
  python -m cl.experiments.paper05_pretrained_jacobian
```

All prior artifacts remain under `results/`; Dataset V2 validation, training, controlled-JVP, and pretrained-JVP directories contain machine-readable aggregates and manifests. Tests cover deterministic replay, no-shortcut information checks, split identity, residual identities, trace parity, local-mask correctness, graph reachability, head-contribution shapes, and zero/replacement equivalence.

F Conclusion

Depth transforms no-shortcut predictive structures into stable correct responses across six controlled formats, often through non-monotone composition and, for nested rules, measurable contextual override. Correct signal grows faster than nuisance-conditioned fluctuation; absolute margin variance need not contract. Nuisance count delays stable decision formation, but pattern-length and span scaling are unsupported. SA windows impose a causal reachability constraint whose satisfaction does not guarantee learned transport. Selective nuisance removal reduces variance without damaging signal, yet indiscriminate truncation harms full-trained models; SA and FF both alter variance and later layers repair much of it. Cumulative JVPs retain local directional information while their norm accuracy deteriorates, a pattern weakly echoed in pinned pretrained Pythia and Qwen. Heads contribute in distributed, non-additive ways. Together these findings turn "layer refinement" into a measurable theory of predictive ensembles, causal cones, information exposure, and interacting residual updates while preserving its failures and boundary conditions.

References

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[3] Stella Biderman et al. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. *ICML*, 2023.

[4] Qwen Team. Qwen3 Technical Report. arXiv:2505.09388, 2025.